



STA 201: Elements of Statistics and Probability

Lecture 10 & 11: Basic Concept of Probability, Conditional Probability & Bayes' Theorem

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Probability

Probability measures the likelihood of occurring an event.

Examples:

- ☐ Weather Forecasting
- ☐ Predicting batting average in cricket matches
- ☐ Politics
- ☐ Choosing appropriate insurance strategies
- ☐ Selecting sport strategies etc.

Basic Terminologies: Three keywords are used while studying probability:

- Experiment
- Outcome
- Event

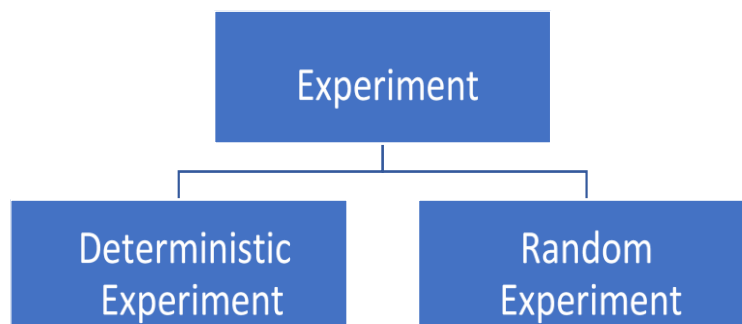
Experiment:

Experiment is an act that can be repeated under given conditions. An experiment is a procedure which can be repeated infinitely but it has a well-defined set of possible outcomes. Unit experiment is called trial.

Example:

- Tossing a coin is an example of experiment since it can be infinitely repeated and it has defined set of possible outcomes which are “heads” & “tails”.
- Rolling a dice is another example of experiment which can be repeated several times and an infinite set of possible outcomes, $S = \{ 1, 2, 3, 4, 5, 6 \}$
- Measuring the lifetime of an automobile is also an example of experiment and its sample space consists of all nonnegative real numbers. That is, $S = [0, \infty)$

Experiment may be Deterministic or predictable and Random or unpredictable.



❑ **Deterministic or predictable experiment:**

An experiment whose outcome is predictable in advanced is called deterministic experiment. Everyone conducting the experiment will get the same result or outcome.

Examples:

- Predicting the amount of money in a bank account if you know the initial deposit and the interest rate.
- The relationship between a circumference and radius of a circle, or the area and radius of a circle.

❑ **Random or unpredictable experiment:**

An experiment whose outcomes can not be predicted with certainty in advance is called random or unpredictable experiment.

Examples:

- An experiment of tossing a coin
- An experiment of rolling a die
- Measuring weight of a person at different times.
- The selection of a numbered of ball (1-50) from an urn.
- Number of defected items produced by a machine by an hour
- Drawing a card from a pack

Outcome:

An outcome is the result of an experiment. In other words, an outcome is a particular result of an experiment. In a tossing coin experiment “Head” is an outcome of the experiment.

Sample Space:

A set or collection of all possible outcomes of a random experiment is called sample space of that random experiment and it is denoted by S. Each outcome of an experiment is a sample point or element in the sample space.

For example,

- Tossing a coin: $S = \{ H, T \}$
- Throwing a dice: $S = \{ 1, 2, 3, 4, 5, 6 \}$
- Lifetime of a lightbulb: $S = \{ x \mid 0 \leq x < \infty \} = [0, \infty)$

- Consider an experiment that consists of rolling two balanced dice, one white and one red are thrown and number of dots on their upper faces are noted, if w be the outcomes of the white die and r be the outcomes of the red die. If we let denote the outcome in which white dice has value w and red dice has value r , then the sample space of this experiment is:

		White Die					
		1	2	3	4	5	6
Red Die	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Event: One or more outcomes of a random experiment constitute an event. An event, E can be defined as a set of outcomes of an experiment or a subset of a sample space (S).

For example, in throwing a die experiment, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.

$E = \{2, 4, 6\}$ is an event, which can be described in words as “the number is even”.

There are several types of events-

Sure event: An event is called sure event when it must occur. The probability of a sure event is one. This means that this kind of event is certain to happen. Probability of everyone who lives will must die will be 1. Because, this is a sure event.

Impossible event: An event is called impossible event when it never happens. This means that probability of an impossible event is always zero and it will never happen. Probability of living without breathing is an impossible event.

Mutually exclusive events:

Two events are called mutually exclusive if both the events cannot occur simultaneously in a single trial. In other words, if one of those events occurs, the other event will not occur.

- In a trial of coin toss experiment, event $E_1 = \{\text{Head}\}$ and event $E_2 = \{\text{tail}\}$ will not occur simultaneously. So, E_1 and E_2 are mutually exclusive events.
- On a day, event $E_1 = \{\text{Rain}\}$ & event $E_2 = \{\text{Sunny}\}$ may occur simultaneously. These are not mutually exclusive events.

Collectively Exhaustive events:

Events of an experiment are said to be collectively exhaustive events if they include all possible outcomes.

- In a coin tossing experiment, $E_1 = \{\text{Head}\}$ and event $E_2 = \{\text{Tail}\}$ are collectively exhaustive, because together they comprise the all the outcomes that are possible in a coin tossing experiment. There are no other possible outcomes of this experiment than these two.

Equally likely events:

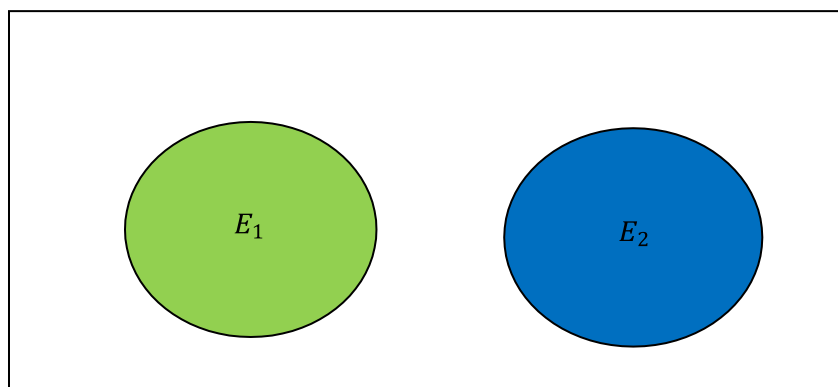
The events of a random experiment are called equally likely if the chance of occurring those events are all equal.

- In a coin to tossing experiment, the events $E_1 = \{\text{Head}\}$ and event $E_2 = \{\text{tail}\}$ are equally likely, because the chance of occurring E_1 is as same as occurring E_2 .
- The sample space of throwing a die is, $S = \{1, 2, 3, 4, 5, 6\}$ and the probability of getting a chosen number = $1/6$. Here, the chance of occurring each numeral is the same and so they are equally likely events.

Disjoint events:

Two events are called disjoint, if they have no common elements between them.

Mutually exclusive events are disjoint events.

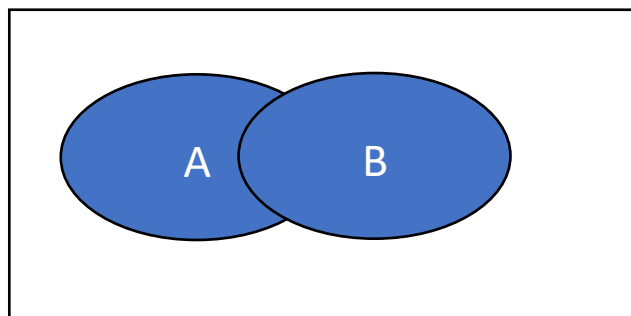


Dependent Events:

If two events have some common elements then both of these events are referred as dependent events.

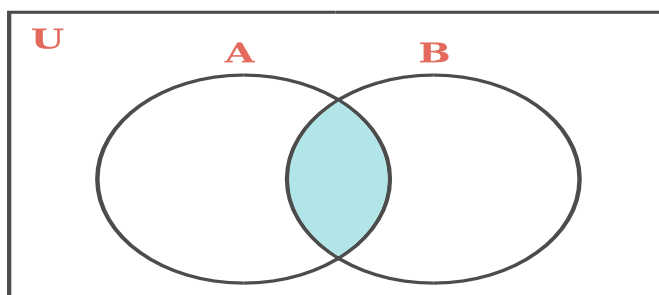
For example, in a deck of 52 cards, if E_1 be an event of selecting 'Red' card and E_2 be another event of selecting 'Queen' card then these events are joint events since there are two 'Queen' cards in a set of 'Red' cards.

- ❑ **Union:** The union of two sets contains all the elements contained in either set (or both sets). The union is notated $A \cup B$, where A and B are two sets.
- ❑ **Intersection:** The intersection of two sets contains only the elements that are in both sets. The intersection is notated $A \cap B$.
- ❑ **Compliment:** The complement of a set A contains everything that is *not* in the set A . The complement is notated A' , or A^c .



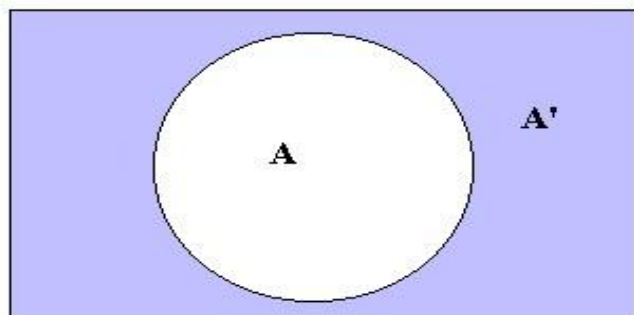
A union B ($A \cup B$)

Elements that belong to either A, B or both



A intersect B ($A \cap B$)

Elements that belong to both A & B



A compliment (A')

Elements that don't belong to A

Independent Events:

Two events are known as independent events if the occurrence of one event does not affect the probability of occurring another event.

For two independent events A and B, the probability that A and B will both occur is found by multiplying the two probabilities.

$$P(A \text{ and } B) = P(A) P(B)$$

Approaches of Assigning Probability

- At first we identify the sample space S of the random experiment.
- We then define our favorable event and assign probability to the event using one of the following 3 basic approaches:
 - ☐ Classical approach
 - ☐ Frequency approach
 - ☐ Subjective approach

Classical approach:

If a random experiment has a total of $n(S)$ possible outcomes, all of which are mutually exclusive, equally likely and collectively exhaustive, such that $n(A)$ of the outcomes are favorable to an event A, then the probability of the event A is defined by,

$$P(A) = \frac{n(A)}{n(S)}$$

Example: Dice throwing experiment -

$$S = \{1, 2, 3, 4, 5, 6\}$$

Consider two events, $E_1 = \{2\}$ and $E_2 = \{2, 4, 6\}$

Here, $n(E_1) = 1$ and $n(E_2) = 3$ and $n(S) = 6$

Therefore, probability of occurring event E_1 is, $P(E_1) = \frac{n(E_1)}{n(S)} = \frac{1}{6}$

Probability of occurring event E_2 is, $P(E_2) = \frac{n(E_2)}{n(S)} = \frac{3}{6} = \frac{1}{2}$

Frequency/ Statistical approach:

If an experiment is repeated n times under the same conditions and event E occurs f times out of n times, then

$$P(E) = \lim_{n \rightarrow \infty} \frac{f}{n}$$

That is, when n is very large, $P(E)$ is very close to the relative frequency of event E .

Example:

In a dice throwing experiment, $S = \{1, 2, 3, 4, 5, 6\}$

And our favorable event is $E = \{2\}$

Let, 2 occurred a total of 998 times out of total 6000 trials.

Therefore,

$$P(E) = \lim_{n \rightarrow \infty} \frac{998}{6000} \approx \frac{1}{6} \quad -$$

Subjective approach:

Based on the judgement (personal experience, prior information and belief etc.), one can assign probability to an event E of a random experiment.

For example; on a day of summer someone made a statement on probability that rain will occur on that day is 0.7, based on his previous experience.

Axioms of Probability

Valid probabilities will follow 3 axioms-

Axiom 1: (Axiom of positivizes): $0 \leq P(E) \leq 1$

Axiom 2: (Axiom of certainty): $P(S) = 1$

Axiom 3: (Axiom of additivity): For a sequence of disjoint events $E_1, E_2, E_3, \dots, E_n$:

$$P(\cup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i)$$

Example 1: A person holds ticket in a lottery that offers 10 prizes and sells 120 tickets. What is the probability that the person will not win a prize?

Solution: Let A be the event of winning a prize.

$$\text{Here, } P(A) = \frac{n(A)}{n(S)} = \frac{10}{120} = \frac{1}{12}$$

$$\text{Thus, } P(A') = 1 - \frac{1}{12} = \frac{11}{12}$$

So, the probability that the person will not win a prize is $\frac{11}{12}$

Example 2: In a community of 400 people, 20 people has a particular disease. If a person is selected randomly from that community, what is the probability that he/ she does not has the disease?

Solution:

Let, D = the randomly selected person has the disease

$$\text{Here, } P(D) = 20/400 = .05$$

$$\therefore P(D') = 1 - .05 = 0.95$$

So, the probability that he/ she does not have the disease is 0.95

Example 3: A dice is thrown in an experiment. What is the probability that an even number will occur?

Solution: The sample space for the experiment is, $S = \{ 1, 2, 3, 4, 5, 6 \}$

Let the event is $E = \{ 2, 4, 6 \}$

Here, $n(E) = 3$ and $n(S) = 6$

Therefore, the probability of occurring the event, E

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = \frac{1}{2} = 0.5$$

Addition Laws

- **For disjoint events A and B-**

The probability that, either event A or event B will occur is,

$$P(A \cup B) = P(A) + P(B)$$

- **For disjoint events A, B, C, ... , and Z**

The probability that, either event A or event B or event C or ... or event Z will occur is,

$$P(A \cup B \cup C \cup \dots \cup Z) = P(A) + P(B) + P(C) + \dots + P(Z)$$

- **For joint events A and B-**

The probability that, either event A or event B or both will occur is,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **For joint events A, B, and C**

The probability that, either event A or event B or event C or any two of them or all will occur is,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

Example 4:

In a company, 60% of the employees have motorcycle, 40% has private car and 20% has both. If an employee is selected randomly from that company, then

- a) What is the probability that the employee has either motorcycle or private car?
- b) What is the probability that the employee has neither motorcycle nor private car?

Solution:

Let, M= the randomly selected employ has motorcycle

C= the randomly selected employee has car

$$\text{Here, } P(M) = \frac{60}{100} = 0.6,$$

$$P(C) = \frac{40}{100} = 0.4$$

$$P(M \cap C) = \frac{20}{100} = 0.2$$

- a) Probability that the person has either motorcycle or private car is,

$$\begin{aligned} P(M \cup C) &= P(M) + P(C) - P(M \cap C) \\ &= 0.6 + 0.4 - 0.2 \\ &= 0.8 \end{aligned}$$

- b) Probability that the person has neither motorcycle nor private car is

$$\begin{aligned} P(M \cup C)^c &= 1 - P(M \cup C) \\ &= 1 - 0.8 \\ &= 0.2 \end{aligned}$$

Conditional probability:

Conditional probability is defined as the likelihood of an event or outcome occurring, based on the occurrence of a previous event or outcome. The probability that event A occurs, given that event B has occurred, is called a conditional probability.

The conditional probability of A, given B, is denoted by the symbol $P(A|B)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}; \text{ for } P(B) > 0$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}; \text{ for } P(A) > 0$$

So, we can write, $P(A \cap B) = P(A|B) P(B)$ (Product Rule)

Or, $P(A \cap B) = P(B|A) P(A)$ ((Product Rule)

Example 5:

In a class of 120 students, 60 are studying English, 50 are studying Mathematics and 20 are studying both. If a student is selected at random from this class, what is the probability that he or she is studying English given that he is studying Mathematics?

Solution:

$$\text{Here, } P(E) = \frac{60}{120} = 0.5$$

$$P(M) = \frac{50}{120} = 0.42$$

$$P(E \cap M) = \frac{20}{120} = 0.17$$

The probability that s/he is studying English given that s/he is studying Mathematics is,

$$P(E|M) = \frac{P(E \cap M)}{P(M)} = \frac{20}{50} = 0.4$$

Multiplication Laws

- **For two dependent events A and B**

The probability that, both event A and event B will occur simultaneously is,

$$P(A \cap B) = P(A|B) P(B)$$

Here, occurrence of event A depends on occurrence of event B.

- **For two independent events A and B**

The probability that, both event A and event B will occur simultaneously is,

$$P(A \cap B) = P(A) P(B)$$

Example 6: In rainy season, it rains 70% of the days in Bangladesh. When it rains, 80% times it makes thunderstorms. What is the probability that, in a particular day of rainy season, it will rain and it will thunderstorm?

Solution: Let, R= it will rain on that particular day

T= it will thunderstorm on that particular day

$$\text{Here, given that, } P(R) = \frac{70}{100} = 0.7 \text{ and}$$

$$P(T|R) = \frac{80}{100} = 0.8$$

Therefore, the probability that, on that particular day of rainy season, it will rain and it will thunderstorm is -

$$\begin{aligned} P(R \cap T) &= P(T|R) P(R) \\ &= 0.8 * 0.7 \\ &= 0.56 \end{aligned}$$

Example 7: A jar contains 3 red, 5 green, 2 blue and 6 yellow marbles. A marble is chosen at random from the jar. After replacing it, a second marble is chosen. What is the probability of choosing a green and then a yellow marble?

Solution: Let, G = Green marble will be chosen

Y = Yellow marble will be chosen

Here, $P(G) = \frac{5}{16}$ & $P(Y) = \frac{6}{16}$

Then, the probability of choosing a green and then a yellow marble is,

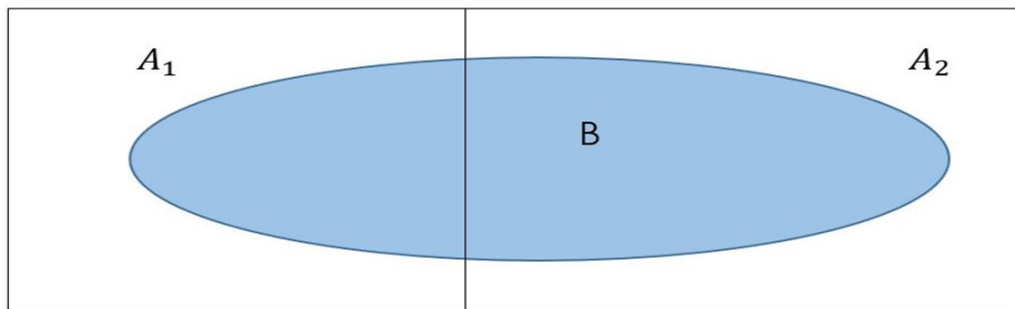
$$P(G \cap Y) = P(G) P(Y)$$

$$= \frac{5}{16} \times \frac{6}{16}$$

$$= \frac{30}{256} = \frac{15}{128}$$

Law of Total Probability: Let, events A_1 and A_2 form partition of S . Let B be an event with $P(B) > 0$.

Then, $P(B) = P(A_1 \cap B) + P(A_2 \cap B) = P(B|A_1) P(A_1) + P(B|A_2) P(A_2)$

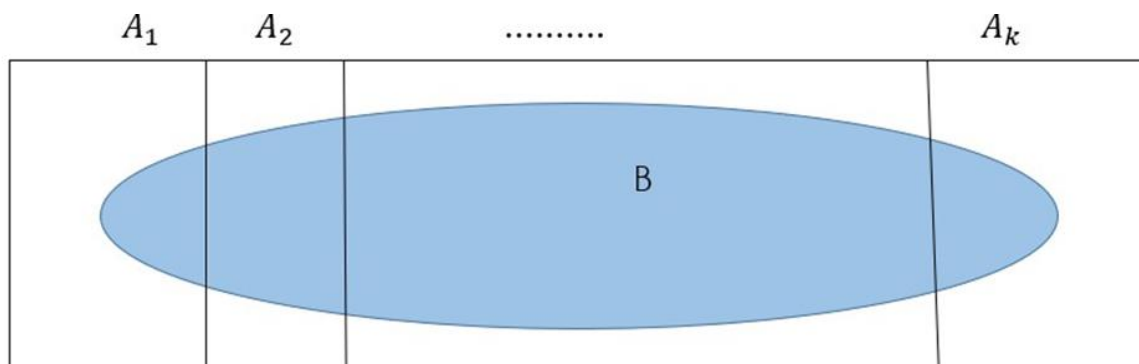


Let, events $A_1, A_2, \dots, A_k$ form partition of S . Let B be an event with $P(B) > 0$. Then,

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_k \cap B)$$

$$= P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + \dots + P(A_k) P(B|A_k)$$

$$= \sum_{i=1}^k P(A_i) P(B|A_i)$$



Example 8: A person has undertaken a mining job. The probabilities of completion of job on time with and without rain are 0.42 and 0.90 respectively. If the probability that it will rain is 0.45, then determine the probability that the mining job will be completed on time.

Solution: Let C = the mining job will be completed on time

R = it rains.

We have, $P(R) = 0.45$,

$$P(\text{no rain}) = P(R') = 1 - P(R) = 1 - 0.45 = 0.55$$

By multiplication law of probability,

$$P(C|R) = 0.42$$

$$P(C|R') = 0.90$$

Since, events R and R' form partitions of the sample space S, by total probability theorem, we have,

$$P(C) = P(C|R) P(R) + P(C|R') P(R')$$

$$= 0.42 \times 0.45 + 0.90 \times 0.55$$

$$= 0.189 + 0.495$$

$$= 0.684$$

So, the probability that the job will be completed on time is 0.684.

Probability using contingency table:

Contingency table:

Contingency table is a power tool in data analysis for comparing categorical variables. Although it is designed for analyzing categorical variables, this approach can also be applied to other discrete variables and even continuous variables.

A general 2×2 contingency table will be like the follows:

X \ Y	Y ₁	Y ₂	Total
X ₁	a	b	a + b
X ₂	c	d	c + d
Total	a + c	b + d	a + b + c + d

Here the two variables are X and Y and each of them have two possible categories.

Example 9.: Suppose a study of speeding violations and drivers who use cell phones produced the following fictional data:

	Speeding violation in the last year	No speeding violation in the last year	Total
Cell phone user	25	280	305
Not a cell phone user	45	405	450
Total	70	685	755

a) Find $P(\text{person is a cell phone user})$.

$$P(\text{person is a cell phone user}) = \frac{\text{Number of cell phone user}}{\text{Total number of persons in the study}} = \frac{305}{755}$$

b) Find $P(\text{person had no violations in the last year})$

$$P(\text{person had no violations in the last year}) = \frac{685}{755}$$

c) Find $P(\text{person is a cell phone user} \mid \text{he had a violation in the last year})$.

$P(\text{person is a cell phone user} \mid \text{he had a violation in the last year})$

$$= \frac{\text{Number of cell phone user who had violation in the last year}}{\text{Total number of persons who has violations in the last year}}$$

$$= \frac{25}{70}$$

Example 10: Below given a contingency table for Smoking status and Cancer status.

Smoking status/ Cancer status	Cancer	Healthy	Total
Smoker	7860	1530	9390
Non-smoker	5390	11580	16970
Total	13250	13110	26360

- What is the probability that a randomly selected person is a smoker
- What is the probability that a randomly selected person has cancer?
- What is the probability that a randomly selected person is both smoker and has cancer?
- If a person is smoker, what is the probability that he also has cancer?

Solution: Let, S= The person is smoker, N= The person is non-smoker

C= The person has cancer, H= The person is healthy

a) The probability that a randomly selected person is a smoker, $P(S) = \frac{9390}{26360} = 0.356$

b) The probability that a randomly selected person is a smoker, $P(C) = \frac{13250}{26360} = 0.503$

c) The probability that a randomly selected person is both smoker and has cancer,

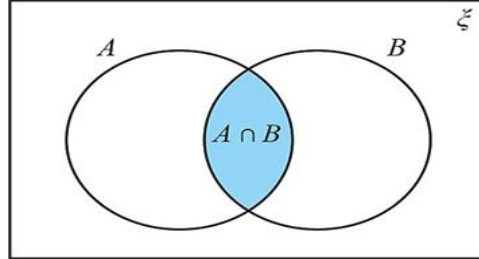
$$P(S \cap C) = \frac{7860}{26360} = 0.298$$

$$d) p(C|S) = \frac{P(C \cap S)}{P(S)} = \frac{\frac{7860}{26360}}{\frac{9390}{26360}} = \frac{7860}{9390} = 0.837$$

Bayes' Theorem

Conditional Probability (Review):

Conditional probability is the probability of one event occurring based on the occurrence of a previous event.



From the diagram above we get two equations:

$$\bullet \quad P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{or,} \quad P(A \cap B) = P(A|B) \cdot P(B) \quad \dots\dots\dots(1.1)$$

$$\bullet \quad P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{or,} \quad P(A \cap B) = P(B|A) \cdot P(A) \quad \dots\dots\dots(1.2)$$

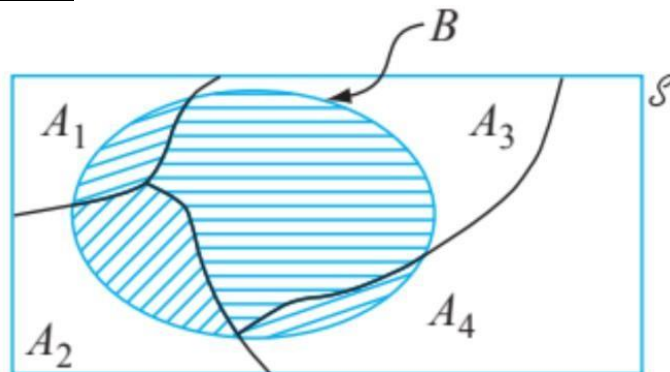
The LHS of both equations 1.1 and 1.2 are equal. Equating them gives:

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$\text{Or,} \quad P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad \dots\dots\dots(1.3)$$

Equation 1.3 represents the equation of Bayes' Theorem

Total Probability (Review):



- Let the rectangle be a sample space
- A_1, A_2, A_3 and A_4 are mutually exclusive and collectively exhaustive events
- Let B be another event in the sample space

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3) + P(B \cap A_4)$$

$$= P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + P(B|A_3) \cdot P(A_3) + P(B|A_4) \cdot P(A_4)$$

$$= \sum_{i=1}^4 P(B|A_i) \cdot P(A_i)$$

In general: $P(B) = \sum^k P(B|A_i) \cdot P(A_i)$

Example 11: 60% of the students in a class are male. 5% of the males and 10% of the females are in the photography club. If a student is randomly selected from the class.

- What is the probability that the student is in photography club?
- If the randomly selected student is in the photography club, what is the chance that the student is male?

Solution: Let M = the student is male

F = the student is female

C = the student is in photography club

$$P(M) = 0.6$$

$$P(F) = 0.4$$

$$P(C|M) = 0.05$$

$$P(C|F) = 0.10$$

$$\begin{aligned} \text{a) } P(C) &= P(C|M) * P(M) + P(C|F) * P(F) \\ &= (0.05*0.6) + (0.10*0.4) = 0.07 \end{aligned}$$

$$\text{b) } P(M|C) = \frac{P(C|M)*P(M)}{P(C)} = \frac{0.05*0.6}{0.07} = 0.43$$

There is 43% chance that a randomly selected student is in the photography club that the student is male.

Example 12: A particular study showed that 12% of men will likely develop prostate cancer at some point in their lives. A man with prostate cancer has 95% chance of a positive test result from a medical screening exam. A man without prostate cancer has a 6% chance of getting a false positive test result. What is the probability that a man has cancer given he has a positive test result?

Solution: Let A_1 = has prostate cancer

A_2 = does not have prostate cancer

B = positive test result

$$P(A_1) = 0.12,$$

$$P(A_2) = 0.88,$$

$$P(B|A_1) = 0.95,$$

$$P(B|A_2) = 0.06$$

$$P(A_1|B) = ?$$

$$P(A_1|B) = \frac{P(B|A_1)*P(A_1)}{P(B)} = \frac{P(B|A_1)*P(A_1)}{P(B|A_1)*P(A_1)+P(B|A_2)*P(A_2)} = \frac{0.95*0.12}{(0.95*0.12)+(0.06*0.88)} = 0.683$$

There is 68.3% chance that a man has cancer given he has a positive test result.

Example 13: A consulting firm submitted a bid for a large consulting contract. The firm's management felt it had a 50-50 chance of landing the project. However, the agency to which the bid was submitted subsequently asked for additional information. Past experiences indicate that for 75% of successful bids and 40% of unsuccessful bids the agency asked for additional information.

- What is the probability of the bid being successful?
- What is the conditional probability of a request for additional information given that the bid will be ultimately successful?
- Compute the probability that the bid will be successful given a request for additional information

Solution:

Let A_1 = the bid is successful

A_2 = the bid is unsuccessful

B = additional information requested

$$P(A_1) = 0.5, \quad P(A_2) = 0.5, \quad P(B|A_1) = 0.75, \quad P(B|A_2) = 0.4$$

a) $P(A_1) = 0.5$

b) $P(B|A_1) = 0.75$

$$c) P(A_1|B) = \frac{P(B|A_1) \cdot P(A_1)}{P(B)} = \frac{P(B|A_1) \cdot P(A_1)}{P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2)} = \frac{0.75 \cdot 0.5}{(0.75 \cdot 0.5) + (0.4 \cdot 0.5)} = 0.652$$

Example 14: Suppose that an insurance company classifies people into one of the three classes – good risks, average risks and bad risks. Their records indicate that the probabilities that good, average, and bad risk persons will be involved in an accident over a 1-year span are 0.05, 0.15 and 0.3 respectively. 20% of the population are "good risks", 50% are "average risks", and 30% are "bad risks".

- What proportion of people have accidents in a fixed year?
- If policy holder A had no accidents in 1987, what is the probability that he or she is a good risk person?

Solution:

Let A_1 = good risk

A_2 = average risk

A_3 = bad risk

B = accident occurs

$$P(A_1) = 0.2$$

$$P(A_2) = 0.5$$

$$P(A_3) = 0.3$$

$$P(B|A_1) = 0.05$$

$$P(B|A_2) = 0.15$$

$$P(B|A_3) = 0.3$$

$$\begin{aligned} \text{a) } P(B) &= P(B|A_1) * P(A_1) + P(B|A_2) * P(A_2) + P(B|A_3) * P(A_3) \\ &= (0.05 * 0.2) + (0.15 * 0.5) + (0.3 * 0.3) \\ &= 0.175 \end{aligned}$$

b) Let B' = accident does not occur

$$\therefore P(B') = 1 - 0.175 = 0.825$$

$$\text{and } P(B'|A_1) = 1 - 0.05 = 0.95$$

$$P(A_1|B') = \frac{P(B'|A_1) * P(A_1)}{P(B')} = \frac{0.95 * 0.2}{0.825} = 0.23$$

If policy holder A had no accidents in 1987, there is a 23% chance that he or she is a good risk.

Example 15: In a bolt factory, machine A, B, C manufacture 25%, 35% and 40% respectively of the total bolts. Of their outputs, 5%, 4% and 2% are respectively, defective bolts. A bolt is drawn at random from the product. If the bolt drawn is found defective, what is the probability that it is manufactured by the machine A or C?

Solution: Follow the class lecture